

Balloon Pop Game Project

Week 4 Progress Report

Project Name: Balloon Pop Game

Week: 4

Development Tool: Apache NetBeans IDE 28

Programming Language: Java

GUI Library: Java Swing

1.Objective:

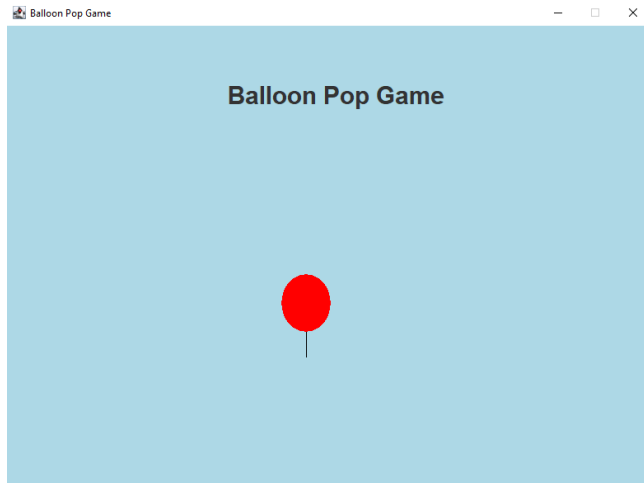
The objective of Week 4 was to add mouse-click detection to the Balloon Pop Game so that the player can click on the moving balloon to pop it. Basic gameplay interaction was also implemented.

2.Work Completed

The following tasks were successfully completed during Week 4:

- 1.Added mouse-click detection to the game.
- 2.Detected the X and Y coordinates of the mouse click.
- 3.Added logic to check whether the click occurred inside the balloon.
- 4.Implemented balloon popping when the player clicks on the balloon.
- 5.Reset the balloon to the bottom after it is popped.
- 6.Generated a new random horizontal position for the balloon.
- 7.Continued the balloon's upward movement after it is popped.
- 8.Kept the balloon movement functionality developed in Week 3.
- 9.Tested the mouse-click and balloon-popping functionality in Apache NetBeans IDE 28.

3.Screenshot of Week 4 GUI:



4.Output:

- 1.After clicking the Start button, the Start button disappears and a red balloon appears near the bottom of the game window.
- 2.The balloon moves upward continuously. The player can click on the balloon while it is moving.
- 3.When the player clicks inside the balloon, the balloon is considered popped. It then returns to the bottom and appears at a new random horizontal position before continuing its movement.
- 4.If the player clicks outside the balloon, nothing happens and the balloon continues moving.

5.Program Logic:

The basic logic of the Week 4 implementation is:

1. The game window is created.
2. The user clicks the Start button.
3. The game starts and the balloon begins moving upward.
4. The program detects mouse clicks on the game window.
5. The X and Y coordinates of the mouse click are obtained.
6. The program checks whether the click is inside the balloon.

7. If the click is inside the balloon, the balloon is popped.
8. The balloon is moved back to the bottom.
9. A new random X-coordinate is generated.
10. The balloon continues moving upward.
11. If the click is outside the balloon, the balloon continues moving normally.

6.Code Logic Explanation:

Code / Concept Logic / Purpose:

- 1.MouseAdapter Provides functionality for handling mouse events.
- 2.MouseEvent Provides information about the mouse action.
- 3.addMouseListener() Adds mouse-click detection to the game window.
- 4.mouseClicked() Executes when the user clicks the mouse.
- 5.e.getX() Gets the X-coordinate of the mouse click.
- 6.e.getY() Gets the Y-coordinate of the mouse click.
- 7.popBalloon() Contains the logic for checking and popping the balloon.
- 8.Ellipse2D.Double Creates an oval-shaped area representing the balloon for click detection.
- 9.balloon.contains(mouseX, mouseY) Checks whether the mouse click occurred inside the balloon.
- 10.balloonY = 500 Moves the popped balloon back to the bottom.
- 11.random.nextInt(700) + 20 Generates a new random horizontal position.
- 12.repaint() Redraws the game window after the balloon position changes.

7.Challenges Faced:

During Week 4, the main challenges were:

- 1.Understanding mouse-click events in Java Swing.
- 2.Understanding how to obtain the X and Y coordinates of a mouse click.
- 3.Understanding how to check whether a click is inside the balloon.

- 4.Connecting the mouse-click event with the balloon-popping method.
- 5.Resetting the balloon after it is clicked.
- 6.Testing the click detection while the balloon was moving.
- 7.These challenges were solved through implementation, testing, and debugging.

8.Learning Outcomes:

After completing Week 4, I learned:

- 1.How to detect mouse clicks using Java Swing.
- 2.How to obtain the coordinates of a mouse click.
- 3.How to check whether a point is inside a graphical object.
- 4.How to create interaction between the player and a game object.
- 5.How to reset a game object after it is clicked.
- 6.How mouse-click detection can be used to create basic gameplay.

9.Conclusion:

Week 4 was completed successfully. Mouse-click detection was added to the Balloon Pop Game, allowing the player to click on the moving balloon and pop it. After being popped, the balloon returns to the bottom at a new random horizontal position and continues moving. This provides the basic interactive gameplay required for the next stage of the project.